

PCB FAB MACHINE — ELECTRONICS ARCHITECTURE

Modular, Two-Board System with Accuracy Enhancements

System Overview

The machine uses only two unique PCB designs:

Main Controller Board (Arduino Uno Q-based)

Universal Tool Board (RP2350B, reused on every tool head)

All tools and future upgrades connect through a shared CAN bus and 24 V power backbone.

The machine is built in modular stages:

Single-sided milling

Flipper upgrade

Cleaning and solder mask

Solder paste

Reflow (manual placement)

Pick-and-place (full automation)

Main Controller Board (Arduino Uno Q)

MCU

- Arduino Uno Q footprint
- SWD/JTAG header
- External crystal for accurate timing

Power

- 24 V input
- Reverse-polarity MOSFET

- Input fuse
- Bulk electrolytic and ceramic capacitors
- 5 V buck converter
- 3.3 V LDO

- Power-good LEDs

Motion

- 4–6 TMC2209 stepper drivers
- UART tuning lines
- DIAG pins for stall detection
- Heatsink and airflow area
- Optical or Hall endstops for X, Y, Z
- Optional encoder inputs

Spindle and Milling

- PWM spindle control
- Spindle enable MOSFET
- RPM tach input

Vacuum Bed and Pneumatics

- MOSFET for vacuum pump
- MOSFETs for valves
- Flyback diodes on all inductive loads

Flipper (added later)

- Stepper outputs
- Endstop sensors
- Optional load cell input

Registration Pins

- Stepper or servo outputs
- Endstop sensors

Safety and Probing

- Hardware E-stop
- Door interlock
- 24 V voltage sense into ADC
- Current sense for spindle, vacuum, and bed
- Ambient temperature sensor

Communications

- CAN bus master
- CAN transceiver
- Jumper-selectable 120 Ω termination
- ESD protection on CAN lines
- USB-C with ESD protection

Debug and Status

- LEDs for 24 V, 5 V, 3.3 V, CAN activity, fault
- Reset button
- User button
- Test pads for all major rails

Universal Tool Board (RP2350B)

Used on every tool head: paste, cleaning, solder mask, laser, via insertion, reflow, pick-and-place, tray feeder, and future tools.

MCU

- RP2350B
- External crystal
- SWD header
- Boot and reset buttons

Power

- 24 V input
- Reverse-polarity MOSFET
- Local fuse
- 5 V buck converter (2–3 A)
- 3.3 V LDO
- Power-good LEDs
- Local 24 V voltage sense

Communications

- CAN transceiver
- Jumper-selectable 120 Ω termination
- Standard 4-pin dock connector (24 V, GND, CANH, CANL)

Motion

- Two TMC2209 stepper driver sockets

- DIAG pins routed
- Stepper connectors

High-Power Outputs

- Eight MOSFET outputs
- Gate resistor, pull-down, flyback diode, test pad on each
- Two relay footprints (optional)

Sensors

- HX711 load cell amplifier
- Load cell connector
- Two MAX31856 thermocouple inputs
- Four analog inputs (filtered)
- Eight digital inputs (pull-ups + RC filters)
- Two servo outputs

Accuracy and Signal Integrity

- RC filters on all analog inputs
- Shielded cable support for load cell and thermocouples
- TVS diodes on CAN, 24 V, thermocouples
- Segregated PCB zones: high-power, analog, digital
- Star-ground layout
- Thick copper pours for MOSFETs and TMCs
- Thermal vias under drivers

Calibration and Identity

- I2C EEPROM storing:
 - Tool Z offset
 - Load cell calibration
 - Thermocouple offsets
 - Tool serial number
 - Tool type
- DIP switch or solder jumpers for hardware ID

Mechanical and Connectors

- Standardized mounting pattern
- Standardized connector layout
- Camera power header (for PnP)
- Optional fan header for hot tools

Tool-by-Tool Usage of the Universal Board

Paste Dispenser

- Stepper driver
- Load cell
- One or two MOSFETs for heater or valve

Cleaning Head

- MOSFETs for brush, IPA pump, flat fan, fine fan

Solder Mask / Slot Coater

- MOSFETs for pump and valve

Laser Head

- MOSFET for laser
- MOSFET for cooling fan

Via Rivet Head

- Load cell
- MOSFET for solenoid press
- MOSFET or stepper for feeder

Reflow Head

- MOSFET for heater
- MOSFET for blower
- MOSFETs for solenoids
- Two thermocouples

Pick-and-Place Head

- Stepper driver for Z
- Stepper driver for rotation
- Load cell
- MOSFETs for vacuum pump, valve, ring light
- Camera powered from board

Tray Feeder

- Stepper driver
- MOSFET for solenoid

- Digital inputs for sensors

System-Level Accuracy Enhancements

Motion Accuracy

- Optical or Hall endstops
- StallGuard collision detection
- Optional encoders on PnP Z and rotation

- Crystal oscillators on both boards

Thermal Accuracy

- Voltage sense on every board
- Current sense on heaters
- Ambient temperature sensor

- Thermocouple cold-junction compensation

Analog Accuracy

- RC filtering on all analog inputs
- External precision reference (2.5 V) for ADC calibration
- Shielded load cell and thermocouple cables
- Star-grounding

- Separate analog and digital ground planes

Calibration Infrastructure

- EEPROM on each tool
- Master reference block on bed for Z drift checks
- Firmware routines for load cell, thermocouple, and Z offset calibration

- Tool identity verification

Noise Immunity

- TVS diodes on CAN, USB, 24 V, thermocouples
- Twisted pair for CAN
- Twisted pair for load cell
- Shielded USB for camera

- Ferrite beads on noisy lines

Safety

- Hardware E-stop
- Door interlock
- Over-current detection

- Over-temperature detection
- Watchdog timers on both MCUs

Modular Upgrade Path

Single-Sided Machine

- Main board
- Spindle
- Vacuum bed

- X/Y/Z motion

Flipper Upgrade

- Flipper motor
- Flipper sensors

- Registration pins

Cleaning and Solder Mask

- Cleaning head
- Slot die coater

- Laser head

Solder Paste

- Paste head

Reflow (Manual Placement)

- Reflow head
- Boards placed by hand, reflow automated

Pick-and-Place

- PnP head
- Tray feeder
- Full automation